KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2024
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1.	a)	What is meant by Natural Language Processing (NLP)? How does it differ from Machine Learning (ML)?	(07)	[CO1]
	b)	Why are NLP tasks sometimes become difficult for a machine to do? Explain with examples.	(12)	[CO2]
	c)	Differentiate between sentence-level classification and token-level classification with suitable examples.	(06)	[CO5]
	d)	Differentiate between linear regression and logistic regression. Which one is appropriate for binary text classification problem? Give a proper justification.	(10)	[CO2]
2.	a)	How does pre-training is related to fine-tuning? Explain.	(07)	[CO1]
	b)	"TF-IDF has better feature representation than BOW."– Is it true? If so, justify.	(08)	[CO3]
	c)	"FFNN is not suitable for processing sequential data." – Explain.	(08)	[CO2]
	d)	Suppose, you need to solve a 3-class text classification problem. How do you perform the task deploying Bi-LSTM? Explain with proper figure.	(12)	[CO5]
3.	a)	Draw vanilla transformer architecture and explain its main parts.	(06)	[CO1]
	b)	What are the main functions of query, key, and value vectors in transformer architecture?	(06)	[CO1]
	c)	Discuss pre-training and fine-tuning procedure of BERT model.	(15)	[CO5]
	d)	"BERT can be used as a generalized understanding model." – Explain.	(08)	[CO1]
4.	a)	How does BERT is utilized to extract features from the text? Explain with detailed diagram.	(12)	[CO5]
	b)	What is sentiment analysis? How can BERT be used to deal with the sentiment analysis task?	(08)	[CO1]
	c)	Discuss machine translation by vanilla transformer model.	(08)	[CO2]
	d)	What is co-reference resolution? Explain with example(s).	(07)	[CO1]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5.	a)	Define corpus. What are the properties of a corpus?	(05)	[CO1]
	b)	"Patterns are greedy." – Why? Design a regular expression to answer the following statement: "Find a computer with at least 6 GHz and 16 GB RAM within 800$."	(10)	[CO1]
	c)	You have two strings "brief" and "drivers". Figure out whether "drive" is closer to "brief" or "drivers" using minimum edit distance algorithm.	(14)	[CO2]
	d)	Define stop words. What are the applications of stop word detection in NLP tasks?	(06)	[CO1]
6.	a)	Define N-gram language model. How can you compute the probability of a sentence of n words P(w), $w = w_1 w_2 \ldots\ldots w_n$? How can you calculate the maximum likelihood by bigram probabilities?	(10)	[CO1]
	b)	Consider the following corpus shown in the following figure. What is the most probable next word?	(15)	[CO3]

<s> I like Henry </s>
<s> I like college </s>
<s> Do Henry like college </s>
<s> Henry I am </s>
<s> Do I like Henry </s>
<s> Do I like college </s>

i) "<s> Do __" applying bigram probability model?
ii) "<s> Do I like __" applying bigram probability model?
iii) "<s> Do I like __" applying trigram probability model?

c) Define smoothing. How can you smooth the language model by interpolation? How to choose the value of 'λ' to maximize the probability? (10) [CO3]

7. a) What is the difference between tokens and embeddings? Explain the pipeline of creating embedding model. (08) [CO4]

b) Consider the following grammar G shown in the following figure. Is the string "The flight includes a meal." in L(G)? Answer it using CYK algorithm. (15) [CO3]

S → NP VP
NP → Det N
VP → V NP
V → includes
Det → the | a
N → meal | flight

c) Given a sequence of ice-cream observations (1 1 3) and an HMM λ = (A,B) in the following figure. Find the best hidden weather sequence Q (like H H H). (12) [CO2]

A		B	
start → HOT	0.25	P(1|HOT) =0.20 P(2|HOT) = 0.35 P(3|HOT) = 0.45	P(1|COLD) =0.50 P(2|COLD) = 0.40 P(3|COLD) = 0.10
start → COLD	0.75		
HOT → HOT	0.60		
HOT → COLD	0.40		
COLD → COLD	0.55		
COLD → HOT	0.45		

8. a) "POS tagging is a disambiguation task" – Justify the statement with example. (10) [CO1]

b) Consider the sentence: "It/PRP would/MD be/VB better/JJR to/TO book/? a/DT ticket/NN via/IN internet/NNP." The word "book" is often used as VB or NN. Given the probabilities below, find the right POS tag for the word "book". (12) [CO3]

P(VB|TO) = 0.67, P(NN|TO) = 0.02, P(book|VB) = 0.50,
P(book|NN) = 0.50, P(DT|VB) = 0.02, P(DT|NN) = 0.06.

c) For Hidden Markov Model (HMM) POS tagging, using the following formula, find the equation of calculating tag transition probabilities. (13) [CO2]

$$\hat{t}_1^n = \underset{t_1^n}{argmax}\, P(t_1^n | w_1^n)$$

CamScanner

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Natural Language Processing

TIME: 3 hours FULL MARKS: 210

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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define corpus. Write a regular expression to find an answer for the following statement. (08) [CO1]
"A computer with 6 GHz and 2 TB storage within 500$".
b) What is lemmatization and stemming? How is lemmatization done? (06) [CO1]
c) How does Byte Pair Encoding (BPE) work for tokenization? Use the following training corpus to tokenize the test word "hugs". (14) [CO1]
"This is the hugging face course. It shows tokenizer algorithms".
d) Make a classification of the types of spelling errors and their solution approaches. (07) [CO1]

2. a) You have two strings *St*1 and *St*2. Develop an algorithm to go from *St*1 to *St*2 with minimum cost of edit. Use insert, delete cost = 1 and update cost = 2. Explain your algorithm using example as *St*1 = "INTENTION" and *St*2 = "EXECUTION". (15) [CO3]
b) Define language model. "Predicting something is difficult". How does *N*-gram language model predict next word? Explain with suitable example. (14) [CO2]
c) How is language model evaluated? (06) [CO2]

3. a) Define text classification. "Naïve Bayes is a form of generative classifier" – justify the statement. (08) [CO3]
b) How does Naïve Bayes learn? Using information from the following table, find the class of the test sentence. (12) [CO3]

	Document	Class
Training	The movie is amazing	Positive
	I do not like the film	Negative
	Great acting and engaging	Positive
	Terrible movie and not recommended	Negative
Test	The acting is superb	?

c) What is parsing in NLP? What are the differences between syntactic and semantic parsing? Explain. (05) [CO3]
d) How does CYK algorithm work? For the following grammar G, is "$baba$" in $L(G)$? (10) [CO3]

$$S \rightarrow AB|BC$$
$$A \rightarrow BA|a$$
$$B \rightarrow CC|b$$
$$C \rightarrow AB|a$$

4. a) "You have a true word w and a noisy channel that modifies w into w'". How can you find the w from w'? Explain. (10) [CO5]
b) How can you define a word as a vector? How can you measure the similarity between two words u and v? Show the similarity of the words from the following table. (15) [CO5]

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

c) What is absolute discounting? Explain what are its advantages. (10) [CO5]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is closed and open class of Parts of Speech (POS)? Explain with example. (05) [CO2]
   b) Explain different type of taggers and describe Hidden Markov Model (HMM) POS tagging using example. (10) [CO1]
   c) Explain Viterbi algorithm for POS tagging with example. (10) [CO1]
   d) How do conditional random fields enhance POS tagging through feature incorporation and why are features challenging to integrate into HMMs? (10) [CO1]

6. a) What is relation extraction? Explain this concept using pattern-based relation extraction and supervised learning algorithm. (10) [CO2]
   b) Explain semi-supervised relation extraction via bootstrapping. (08) [CO2]
   c) What are temporal lexical triggers? How do we detect events? (06) [CO2]
   d) How is the *tf-idf* score computed for terms in documents and how can it be used to calculate the cosine similarity between a query and documents using BM25? Demonstrate for the following dataset. (11) [CO5]

   Assume, $|d| = 4$, $|d_{avg}| = 3$, $k = 1.3$, $b = 0.75$.

   Query: sweet love
   Doc 1: Sweet sweet nurse! Love?
   Doc 2: Sweet sorrow
   Doc 3: How sweet is love?
   Doc 4: Nurse!

7. a) Explain the evaluation of information retrieval systems and describe IR-based factoid question answering with a diagram. (08) [CO5]
   b) How can BERT-based models be effectively trained and evaluated for answer span extraction in information retrieval-based question answering systems? (09) [CO5]
   c) What is entity linking? Explain the process of entity mention detection using mathematical formulas and a diagram. (08) [CO5]
   d) What is knowledge-based question answering? Describe Watson's question answering system with the help of a diagram. (10) [CO5]

8. a) How does attention mechanism work in machine translation and why is it important for improving translation quality? (09) [CO4]
   b) Explain hourglass metaphor in text-to-speech with proper diagram. (08) [CO4]
   c) What is the difference between tokens and embeddings? Explain the pipeline of creating embedding model. (08) [CO4]
   d) Why is beam search necessary? Explain beam search with example. (10) [CO2]

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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define corpus. What are the properties of a corpus? (05) [CO1]
   b) "Patterns are greedy" – why? Design a regular expression to answer the following statement: "Find a computer with at least 6 GHz and 512 GB RAM for less than BDT 4000". (10) [CO1]
   c) Define subword tokenization. Explain the BPE algorithm for subword tokenization. Use the following corpus to explain the algorithm.
   "low low low low low lowest lowest Newer Newer Newer Newer wider wider wider new new"
   Test Set: newer, lower. (15) [CO2]
   d) Define stop words. What are the applications of stop word detection in NLP tasks? (05) [CO1]

2. a) You have two strings "brief" and "drivers". Figure out whether "drive" is closer to "brief" or "drivers" using minimum edit distance algorithm. (14) [CO2]
   b) What is *n*-gram language model? Find the uni-gram, bi-gram, and tri-gram from the following corpus.
   "This is Big Data AI Book". (06) [CO3]
   c) Why do you need smoothing techniques for *N* gram model? Explain the following smoothing techniques: (15) [CO3]
      i) Add-1 smoothing.
      ii) Stupid back off.
      iii) Kenser-Ney smoothing.

3. a) Give a formal definition of supervised text classification. How can you differentiate generative and discriminative classifier? (05) [CO4]
   b) What is "Bag of words" model? What are the assumptions of Bag of words model? How does Bag of words applied for multinomial Naïve Bayes Classifier? Using the following training set find the class of the test set. (15) [CO4]

		Document	Class
Training	:	Just plain boring	–ve
		No surprise and few laughs	–ve
		Very powerful	+ve
		Most fun film of the summer	+ve
Test	:	Predictable with no fun	?

   c) When is "Accuracy" not a good metric? Explain with example. (07) [CO3]
   d) "The noisy channel is a kind of Bayesian inference" – explain the term. (08) [CO3]

4. a) What are the types of spelling errors? you wanted to type 'three' but mistakenly you typed 'there'. How can you detect such an error automatically? How to correct these types of errors? (12) [CO3]
   b) What are the common structural ambiguities of Natural Language Processing? Explain. (06) [CO3]
   c) What is partial parsing? What are the applications of partial parsing? (05) [CO3]
   d) Consider the following grammar G. Using the CYK algorithm, find whether "ababa" is L(G) or not. (12) [CO3]

$$S \rightarrow AB|BC$$
$$A \rightarrow BA|a$$
$$B \rightarrow CC|b$$
$$C \rightarrow AB|a$$

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the large language model (LLM) with an example. Discuss computational resources and energy consumption in LLM. (10) [CO1]
   b) "Parts-of-Speech (POS) tagging is a disambiguation task" – justify the statement with example. (10) [CO2]
   c) Discuss POS tagging and tagsets for English. (10) [CO3]
   d) How difficult is POS tagging in English? (05) [CO3]

6. a) Discuss Conditional Random Fields using Bayes' rule. (10) [CO2]
   b) We have a dataset (NewG). Some information about the dataset is given below. (25) [CO2]

No. of VB	:	17619	Starting probability VB	:	0.024
No. of MD	:	11338	Starting probability MD	:	0.11
No. of NN	:	16559	Starting probability NN	:	0.86
$t_{i-1} - t_i$			$t_i - w_i$		
VB – VB	:	8583	VB – race	:	5391
VB – MD	:	6988	VB – will	:	6308
VB – NN	:	5826	VB – begin	:	7488
MD – VB	:	7984	MD – race	:	6557
MD – MD	:	4204	MD – will	:	6635
MD – NN	:	4046	MD – begin	:	6905
NN – MD	:	5006	NN – race	:	4163
NN – NN	:	4568	NN – will	:	6103
			NN – begin	:	7863

Use the dataset to answer the following questions:

i) Calculate the transition probability (A).
ii) Calculate the emission probability (B) for sentence: "race will begin".
iii) Find the tag sequence for the sentence "race will begin" using hidden Markov model.

7. a) Discuss the sequence to sequence model with attention mechanism (AM). Demonstrate the limitations without AM. (10) [CO2]
   b) Can hidden Markov models incorporate features? (05) [CO2]
   c) What is bootstrapping in information extraction? Explain the architecture for bootstrapping. (10) [CO4]
   d) How do we detect events? Write down features commonly used in both rule-based and statistical approaches to event detection. (10) [CO4]

8. a) Explain term weighting and document scoring. (05) [CO5]
   b) (12) [CO5]

Query	:	Hello dear
doc1	:	Hello my friends
doc2	:	Hello Karim, hello Rahim
doc3	:	My dear friend, Hello
doc4	:	Hello, Dear, dear

Using term weighting and document scoring, rank the document for Query.

   c) Discuss language models (LMs) to do question answering (QA). Write down the comparative analysis between LM QA models and classic QA models. (10) [CO5]
   d) What is homograph disambiguation? What problems does CMU face in this regard, and how does UNISYN address these issues? (08) [CO4]

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the differences between Natural Language Processing (NLP) and Text Mining? Explain. (06)
   b) What is Lemmatization and Stemming? Discuss one popular Lemmatization and Stemming algorithm with example. (10)
   c) Develop the regular expression to find all the instances of the word "there" in a text. (07)
   d) You have two strings X of length n and Y of length m. Develop an algorithm to edit the string X to Y. How many operations you need? Explain your algorithm for X = "KITTEN" and Y = "SITTIN". (12)

2. a) Define N-gram language model. How can you compute the probability of a sentence of n words P(w), $w = w_1 w_2 \dots w_n$. How can you calculate maximum likelihood by bigram probabilities? (10)
   b) Consider the following corpus. (15)

<S> I Like Henry </S>
<S> I Like College </S>
<S> Do Henry Like College </S>
<S> Henry I am </S>
<S> Do I Like Henry </S>
<S> Do I Like College</S>

What is the most probable next word
   i) "<S> Do —" applying bigram probability model?
   ii) "<S> Do I Like —" applying bigram probability model?
   iii) "<S> Do I Like —" applying trigram probability model?

   c) Define smoothing. How can you smooth the language model by interpolation? How to choose the value of 'λ' to maximize the probability? (10)

3. a) What is Bag of words representation? How do you apply Bag of words assumption and conditional independence for naïve bayes classifier? Explain. (12)
   b) Consider the following Training dataset. (08)

item	class
Just plains boring	–
Entirely Predictable and lacks energy	–
No surprises and very few laughs	–
Very powerful	+
The most fun film of the summer	+

Find the class of the following sentence "Predictable with no fun", using naïve bayes classifier.
   c) How can you apply noisy channel for real world spelling correction? (10)
   d) "Accuracy is not a good metric when the goal is to discover something rare" –justify the statement. (05)

4. a) What are the common structural ambiguities for any NLP task? Give example(s). (08)
   b) Consider the following grammar G (13)

$S \rightarrow NP\ VP$
$NP \rightarrow Det\ N$
$VP \rightarrow V\ NP$
$V \rightarrow includes$
Det → the | a
$N \rightarrow meal \mid flight$

Is the string "The flight includes a meal." in L(G)? Answer it using CYK algorithm.

c) Define PCFG. What is the condition to be maintained by a PCFG? How can you disambiguate the parsing using PCFG? (09)

d) What are the applications of shallow parsing? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is closed class and open class of parts of speech? Explain with example. (08)

b) Give a short description of Viterbi algorithm. What are the applications of the algorithm? (07)

c) Use the following table to answer the following questions. (20)

We have a dataset (NextD). Some information about the dataset is given below.	
No. of VB: 10756	Starting probability VB: 0.44417
No. of MD: 19375	Starting probability MD: 0.7836
No. of NN: 14479	Starting probability NN: 0.25762
$(t_{i-1} - t_i)$	
VB–MD: 9544	VB–race: 6272
VB–NN: 9385	VB–will: 8977
MD–VB: 9918	VB–begin: 5569
MD–NN: 9590	MD–race: 6891
NN–VB: 9065	MD–will: 6301
NN–MD: 9754	MD–begin: 4857
VB–VB: 0	NN–race: 4910
NN–NN: 0	NN–will: 4012
MD–MD: 0	NN–begin: 4617

i) Calculate the transition probability (A).

ii) Calculate the emission probability (B) for sentence: "race will begin".

iii) Find the tag sequence for the sentence "race will begin" using Viterbi algorithm.

6. a) Define Named Entity Recognition. Why recognizing name entity is difficult? (07)

b) Discuss conditional random fields (CRF). Write down the features in a CRF PoS tagger. (08)

c) Discuss the IR-based Factoid Question Answering system with proper diagram(s). (10)

d) Define knowledge-based question answering. Discuss four broad stages of Watson QA with block diagram(s). (10)

7. a) Write down classes of algorithms for relation extraction. Describe Relation Extraction via Supervised Learning and Feature-based supervised relation classifiers. (10)

b) Write down three ways to extract Times and temporal Expression. Explain with examples. (08)

c) Discuss the encoder-decoder model for machine translation with proper figures. (10)

d) Define entity linking. Discuss neural graph-based linking with proper diagram(s). (07)

8. a) Discuss the information Retrieval question-answering system. Explain with examples. (08)

b) Consider the following data: (12)

Query	:	Sweet love
Doc 1	:	Sweet sweet nurse! Love?
Doc 2	:	Sweet sorrow
Doc 3	:	How sweet is love?
Doc 4	:	Nurse!

Using Term Weighting and document scoring, Rank the document for query.

c) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (07)

d) Speech synthesis performs text-to-waveform mapping in two steps. Name and discuss the steps using Hourglass metaphor. (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
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Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) "Pattern matching by regular expression is greedy" – Explain the term taking example to match the word 'There' from a string. (10)
   b) Given two strings $S1$ and $S2$ and three operations ADD, DELETE and EDIT. How many minimum operations are needed to convert $S1$ to $S2$? Derive the algorithm to answer the question. Apply the algorithm for $S1$ = "*abcdef*" and $S2$ = "*azced*". If each of the operation costs 1 unit, find the total cost to convert $S1$ to $S2$. (20)

2. a) "Minimizing perplexity is same as maximizing the probability" – Explain the term. (12)
   b) Consider the following corpus: (12)

   $\langle S\rangle abc\langle /S\rangle$
   $\langle S\rangle bac\langle /S\rangle$
   $\langle S\rangle abc\langle /S\rangle$

   Calculate the bigram probabilities $P(a|\langle S\rangle)$, $P(c|b)$ and $P(b|a)$.
   c) How do you handle the stop words for text classification? (06)

3. a) Given the following text classification for C_1 and C_2 (14)

   $PQR \rightarrow C_1$
   $PPS \rightarrow C_1$
   $PT \rightarrow C_1$
   $UVP \rightarrow C_2$

   Compute the most likely class for the text "*PPPUV*".
   b) Consider the following grammar in CNF (16)

   $S \rightarrow AB \mid BC$
   $A \rightarrow BA \mid a$
   $B \rightarrow CC \mid b$
   $C \rightarrow AB \mid a$

   Is '*baaba*' in $L(G)$? Explain your answer using CYK algorithm.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) "POS tagging is a disambiguation task" – Justify the statement with example. (10)
   b) Consider the sentence: (08)
   "*It/PRP would/MD be/VB better/JJR to/TO book/?a/DT ticket/NN via/IN internet/NNP*".
   The word "*book*" is often used as *VB* or *NN*. Given the probabilities below, find the right POS tag for the word "race".

   $P(VB|TO) = 0.67$, $P(NN|TO) = 0.02$, $P(book|VB) = 0.50$,
   $P(book|NN) = 0.50$, $P(DT|VB) = 0.02$, $P(DT|NN) = 0.06$.

   c) What is prosody? Discuss about the three aspects of prosody. Discuss the effect of zero situation. (12)

5. a) Given a sequence of ice-cream observations (1 1 3) and an HMM $\lambda = (A, B)$ in the following figure, find the best hidden weather sequence $Q(like\ H\ H\ H)$. (12)

A		B	
$Start \rightarrow HOT$	0.25		
$Start \rightarrow COLD$	0.75	$P(1 \mid HOT) = 0.2$	$P(1 \mid COLD) = 0.5$
$HOT \rightarrow HOT$	0.60	$P(2 \mid HOT) = 0.35$	$P(2 \mid COLD) = 0.4$
$HOT \rightarrow COLD$	0.40	$P(3 \mid HOT) = 0.45$	$P(3 \mid COLD) = 0.1$
$COLD \rightarrow COLD$	0.55		
$COLD \rightarrow HOT$	0.45		

b) Define the term odds for logistic regression. Show that the observation should be labeled true if $\sum_{i=0}^{N} w_i f_i > 0$. (12)

c) What are the components of Markov chain? (06)

6. a) Discuss about the steps of text to waveform mapping in speech synthesis using Hourglass metaphor. (10)

b) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (10)

c) What are the use of Question Answering System (QAS)? Discuss about the main phases of QAS architecture. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Why is the pattern matching by regular expression greedy? Explain with example. (09)
   b) Define text normalization. What is the difference between lemmatization and stemming? Explain with example. (10)
   c) Explain porter's algorithm with example. (11)
   d) What is types and tokens? How many types and tokens are there in the following sentence? "He stepped out into the hall, was delighted to encounter a water brother." (05)

2. a) "Finding the minimum edit distance is problem of the category 'dynamic programming' "- Explain the statement to convert one String X of length n to a String Y of length m. (12)
   b) Apply the minimum edit distance algorithm (using insertion cost = 1, deletion cost = 1 and substitution cost = 2) of "leda" to "deal". Show your work in edit distance grid. (10)
   c) Define bag of words problem. How is bag of words conditional independence assumption applied in Naïve Bayes text classification? Explain. (13)

3. a) Define Stop word. How do you handle Stop word in text classification? (08)
   b) Given the following text classification for classes C_1 and C_2 (10)
      i) PQR $\rightarrow C_1$
      ii) PPS $\rightarrow C_1$
      iii) PT $\rightarrow C_1$
      iv) UVP $\rightarrow C_2$

      Compute the most likely class for the text "PPPUV".
   c) Consider the following likelihood for a review (05)

	Pos	Neg
I	0.09	0.16
always	0.07	0.06
like	0.29	0.06
foreign	0.04	0.15
films	0.08	0.11

   What class will the Naïve Bayes assign to the sentence "I always like foreign films"?
   d) Find the context free rules and hence the context free grammar (CFG) for the following English sentences. (12)
      i) I want a morning flight.
      ii) Which flight serves breakfast?
      iii) Show the lowest fare.
      iv) The flight should be at 11 am.

4. a) Consider the following grammar in CNF (10)

$$S \rightarrow AB \mid BB$$
$$A \rightarrow CC \mid AB \mid a$$
$$B \rightarrow BB \mid CA \mid b$$
$$C \rightarrow BA \mid AA \mid b$$

   Is 'aabb' in L(G)? Justify your answer using CYK algorithm.
   b) How does Early Parser differ from CYK Parser? Explain finite state chunking with example. (10)
   c) Define PCFG? How can you disambiguate a syntactic Parsing by PCFG? (06)
   d) Explain the IR based factoid question answering using example. (09)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define Natural Language Processing (NLP). Why do we need NLP? Write some industrial applications of NLP? Derive (10)
   b) What does smoothing mean? ~~Drive~~ the equation for Laplace smoothing. (10)
   c) Given the following corpus: (07)

   I like NLP.
   I like Deep Learning.
   NLP and Deep Learning have huge area for research.

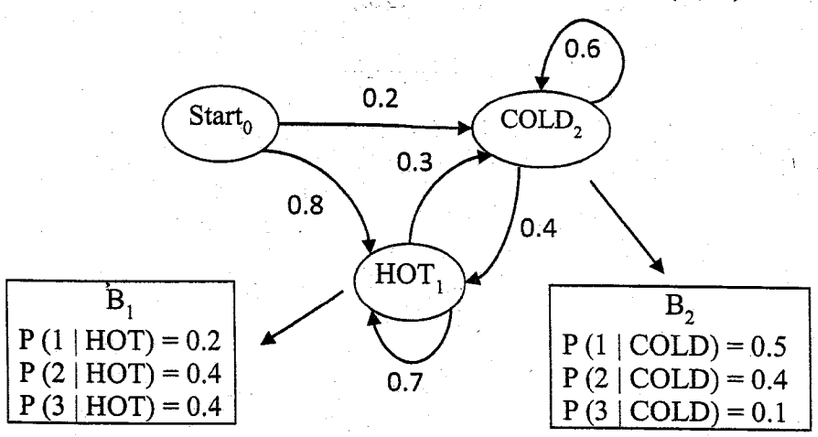
   Using Bigram Language model with add-one smoothing, what is P (NLP | like)?
   d) Define zero situation and discuss the effect of it. (08)

6. a) "POS tagging is a disambiguation task" – Justify the statement with example. (08)
   b) Define perplexity. Derive the equation of computing W with a Bigram Language model. (10)
   c) For Hidden Markov Model (HMM) Pos tagging, using the following formula, find the equation of calculating tag transition probabilities (10)

$$\hat{t}_1^n = \underset{t_1^n}{argmax}\, P\,(t_1^n \mid w_1^n)$$

   d) Consider the sentence: "Eight/CD horses/NNS will/MD race/? For/IN the/DT cup./NN." Given the probabilities below, find the right Pos tag for the word "race".(VB or NN). (07)
   P (VB | MD) = 0.0045, P (NN | MD) = 0.062, P (race | NN) = 0.048, P (race | VB) = 0.014, P (IN | VB) = 0.0012, P (IN | NN) = 0.0024

7. a) Define Markov chain. Explain the components of Markov chain. (06)
   b) For logistic regression, show that P (y = true | x) = 1 | (1 + $e^{-w.f}$) (10)
   c) Given a sequence of ice-cream observations 3 1 3 and an HMM λ = (A, B) in the following figure. (12)



   Find the best hidden weather sequence Q (like H H H).
   d) What are the statistical models for processing text and speech? Give example of fully connected & left to right HMM. (07)

8. a) Define sentence tokenization. Draw the architecture of a TTS. (07)
   b) Discuss about the steps of speech synthesis (text to waveform). Show the Hourglass Metaphor. (10)
   c) What is prosody? Discuss about the three aspects of prosody. (10)
   d) What is accent ratio? Why is accent ratio important? (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2018
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Disjunction, Grouping and Precedence for pattern matching in regular expressions? Explain with example. (10)
   b) "Pattern matching by regular expressions are greedy." – Justify the statement. (09)
   c) Design a regular expression to find all instances of the word "the" in a text. (10)
   d) Define types and tokens. How many types and tokens are there in the following sentence: (06)
   "They picnicked by the pool, then lay back on the grass and looked at the stars."

2. a) What is Lemmatization and Stemming? How is Lemmatization done? (08)
   b) What are the operations for editing one string to another? Explain. (06)
   c) Explain the algorithm to edit one string X of length n to a string Y of length m. Show the steps of your algorithm for X = INTENTION and Y = EXECUTION. (12)
   d) Discuss about the problem with Maximum Likelihood. How does the Laplace (add-1) Smoothing solve the problem? (09)

3. a) "Accuracy is not a good metric when the goal is to discover something that is rare." – Justify the statement with example. Propose a metric to solve the drawbacks of accuracy. (10)
   b) Given the following short movie reviews, each labeled with a genere, either comedy or action: (10)
      i) fun, couple, love, love ⇒ comedy
      ii) fast, furious, shoot ⇒ action
      iii) couple, fly, fast, fun, fun ⇒ comedy
      iv) furious, shoot, shoot, fun ⇒ action
      v) fly, fast, shoot, love ⇒ action

   Consider a new document D: fast, couple, shoot, fly. Compute the most likely class for D.
   c) Find the context free rules and hence the Context Free Grammar (CFG) for the following English sentences: (15)
      i) I want a morning flight.
      ii) I want a flight from Ontario to Chicago.
      iii) Show me the cheapest fare that has lunch.
      iv) Do any of these flights have stops?
      v) Which flights serves breakfast?

4. a) Consider the following grammar in CNF. (10)

$$S \rightarrow AB \mid BC$$
$$A \rightarrow BA \mid a$$
$$B \rightarrow CC \mid b$$
$$C \rightarrow AB \mid a$$

   Is '*baaba*' in L(G)? Explain your answer using CYK algorithm.
   b) Define shallow parsing. What are the applications of shallow parsing? (05)
   c) Define Probabilistic Context Free Grammar (PCFG). Consider the following PCFG. (12)

$$S \rightarrow NPVP \mid AuxNPVP \mid VP \; [0.8 \mid 0.1 \mid 0.1]$$
$$NP \rightarrow Pronoun \mid Proper\text{-}noun \mid DetNominal \; [0.2 \mid 0.2 \mid 0.6]$$
$$Nominal \rightarrow Noun \mid NominalNoun \mid NominalPP \; [0.3 \mid 0.2 \mid 0.5]$$
$$VP \rightarrow verb \mid verbNP \mid VPPP \; [0.2 \mid 0.5 \mid 0.3]$$
$$PP \rightarrow PrepNP \; [1.0]$$
$$Det \rightarrow the \mid a \mid that \mid this \; [0.6 \mid 0.2 \mid 0.1 \mid 0.1]$$
$$Noun \rightarrow book \mid flight \mid meal \mid money \; [0.1 \mid 0.5 \mid 0.2 \mid 0.2]$$
$$verb \rightarrow book \mid include \mid prefer \; [0.5 \mid 0.2 \mid 0.3]$$
$$Pronoun \rightarrow I \mid he \mid she \mid me \; [0.5 \mid 0.1 \mid 0.1 \mid 0.3]$$
$$Proper\text{-}noun \rightarrow Houston \mid NWA \; [0.8 \mid 0.2]$$
$$Prep \rightarrow from \mid to \mid on \mid near \mid through \; [0.25 \mid 0.25 \mid 0.1 \mid 0.2 \mid 0.2]$$

      i) Find the probability of the sentence "book the flight through Houston".
      ii) Using the disambiguation algorithm select the proper parse tree.
   d) What are the stages of IR based question answering? Explain. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define Natural Language Processing (NLP). What are the major areas of research and development of NLP? (10)

   b) What does *n*-gram mean? Drive the equation of calculating the probability for *n*-grams model. (10)

   c) Consider the following corpus. (08)

   <s> I am Sam </s>
   <s> Sam I am </s>
   <s> I am Sam </s>
   <s> I do not like green eggs and Sam </s>

   Using a Bigram Language model with add-one smoothing, what is $P(sam \mid am)$? Include <s> and </s> in your counts just like any other token.

   d) What is absolute discounting? What is its advantages? (07)

6. a) What is closed class and open class of Part-of-Speech (POS)? Explain with example. (08)

   b) Discuss about Rule-Based POS tagging. Write the ADVERBIAL-THAT RULE. (12)

   c) For Hidden Markov Model (HMM) POS Tagging, using the following formula, find the equation of calculating tag transition probabilities. (08)

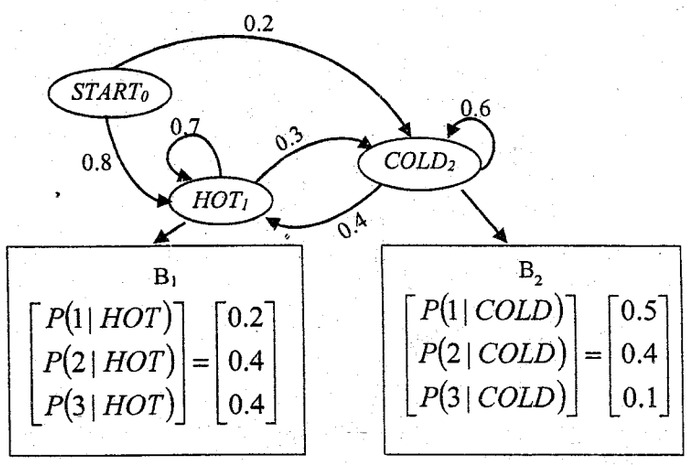
$$\hat{t}_1^n = \arg\max_{t_1^n} P(t_1^n \mid w_1^n)$$

   d) Consider the sentence: "Secretariat/NNP is/BEZ expected/VBN to/TO race/? Tomorrow/NR". The word "race" is often used as VB or NN. Given the probabilities below, find the right POS tag for the word "race". (07)

$$P(NN \mid TO) = 0.00047, P(VB \mid TO) = 0.83, P(race \mid NN) = 0.00057, P(race \mid VB) = 0.00012,$$
$$P(NN \mid VB) = 0.0027, P(NR \mid NN) = 0.0012.$$

7. a) HMM characterized by three fundamental problems. Name and discuss about the problems. (09)

   b) Given a sequence of ice-cream observations 313 and an HMM $\lambda = (A, B)$ in the following figure, find the best hidden weather sequence $Q(like\ H\ H\ H)$. (12)



   c) Define the term odds for logistic regression. Show that the observation should be labeled true if $\sum_{i=0}^{N} w_i f_i > 0$. (09)

   d) Write the three-steps of Forward Algorithm. (05)

8. a) Name and discuss about the types of TTS. (06)

   b) Speech Synthesis perform text to waveform mapping in two-steps. Name and discuss about the steps. Using Hourglass Metaphor. (12)

   c) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (10)

   d) Define text normalization. Why does text normalization important for Speech Synthesis? (07)